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INFLUENCE OF OXYGEN ON PERMEATION OF HYDROGEN ISOTOPES THROUGH GROUP 5 METALS

Yuji Hatano[1*], Andrei Busnyuk[2], Vasily Alimov[2], Alexander Livshits[2], Yukio Nakamura[3], Masao Matsuyama[1]

[1]*Hydrogen Isotope Research Center, University of Toyama, Gofuku 3190, Toyam 930-8555, Japan*
[2]*Bonch-Bruyevich University, 61 Moika, St. Petersberg 191186, Russia*
[3]*National Institute for Fusion Science, Toki 509-5292, Japan*
** hatano@ctg.u-toyama.ac.jp*

Group 5 metals (V, Nb and Ta) are potential candidates of tube material in vacuum permeator for tritium recovery from Pb-17Li liquid blanket system. From this viewpoint, the influence of oxygen on the surface reaction rates of hydrogen on V and Ta were examined in an ultra-high vacuum apparatuses at elevated temperatures, and the results were compared with the data on Nb acquired in a previous study. The surface reaction rates of hydrogen on V and Ta, and consequently permeation rates, decreased with increasing oxygen concentration in the bulk as previously observed for Nb. These observations were ascribed to the increase in surface oxygen coverage with increasing bulk oxygen concentration. The weakest influence of oxygen on hydrogen permeation rate was observed for V. The expected permeation rate through V under typical blanket conditions, however, was not necessarily high due to high oxygen solubility in V. The evaluation indicated that the highest permeation rate should be obtained with Nb under typical blanket conditions.

I. INTRODUCTION

Tritium recovery by vacuum permeation is proposed in dual coolant Pb–17Li breeder blanket concept.[1] Group 5 metals (V, Nb and Ta) are potential candidates of tube material for the permeator because of high permeability of hydrogen isotopes and sufficient compatibility with Pb–17Li.[1] Terai et al.[2] have examined the feasibility of an Nb permeation window for tritium recovery for this type of liquid breeder material and reported that permeation rate of deuterium was much smaller than that expected from diffusivity and solubility of deuterium in Nb. They ascribed this observation to the growth of oxide films on Nb surfaces.[2]

The growth of oxide films on metal surfaces could be avoided at sufficiently high temperature and low oxygen potential due to dissolution of oxygen into the bulk.[3] It, however, does not necessarily mean that clean surfaces are obtained; the metal surfaces under such conditions are covered by segregated nonmetallic impurities such as oxygen. It is known that the reactivity of metal surfaces covered by segregated impurity varies with the bulk concentration of corresponding impurity because of the dependence of surface impurity coverage on the bulk concentration.[3,4] The reduction in surface reaction rates of hydrocarbons with increasing bulk oxygen concentration has been observed for Nb and Ta by Hörz and Fromm[3] and Hörz et al.[4] at temperatures higher than 1000 ^{O}C.

The present authors have examined the influence of oxygen on the permeation of hydrogen through Nb under the conditions relevant to the blanket concept and observed the sharp reduction in permeation rate with increasing bulk oxygen concentration.[5] Nevertheless, serious deterioration in tritium recovery rate was not expected at the bulk concentration under equilibrium with the oxygen potential in Pb-17Li. The influence of oxygen on the permeation through V and Ta, however, has not been investigated.

In the present study, the variation of surface reaction rates of hydrogen on V and Ta with bulk oxygen concentration was examined by permeation and thermal desorption techniques. Then the suitability of these metals for tritium recovery was compared with that of Nb.

II. EXPERIMENTAL

The influence of oxygen on the surface reaction of H_2 on V was examined by permeation experiments carried out in an ultra-high vacuum (UHV) apparatus in the manner described elsewhere.[5-7] A membrane of polycrystalline V (0.1 mm thickness) was first sputter-cleaned by hydrogen ions at elevated temperatures (600-1000 ^{O}C) to remove impurities from not only the surface

but also the bulk by continuous surface segregation. Then a known amount of oxygen was introduced into the specimen by exposure to O_2 gas at around 600 ^{O}C. The permeation of H_2 was examined in a temperature range from 180 to 1200 ^{O}C at upstream H_2 pressures from 1×10^{-3} to 5 Pa.

In the case of Ta, the influence of oxygen on the surface reaction rate of hydrogen was examined by the thermal desorption technique described in Ref. 8. The specimen used was a thin sheet of polycrystalline Ta (180×8×0.02 mm). Impurities in the specimen were removed by heating in UHV at 2200 ^{O}C, before a known amount of oxygen was introduced by exposure to O_2 gas at 650 ^{O}C. After exposing the specimen to atomic hydrogen at 200 ^{O}C, thermal desorption of H_2 was examined by increasing the specimen temperature to 850 ^{O}C in step-by-step manner. The rate constant for recombinative release, $k_r(C_O, T)$, at given bulk oxygen concentration, C_O, and temperature, T, was evaluated from desorption flux, ϕ_{des}, and hydrogen concentration, C_H, in the specimen using the following correlation: $2\phi_{des} = k_r(C_O, T)C_H^2$.

The permeation probability, i.e. the fraction of permeation flux, ϕ_P, to the incident H_2 flux onto the upstream surface, ϕ_{inc}, was evaluated from the values of $k_r(C_O, T)$. If the properties of upstream and downstream surfaces of a tube are identical to each other, the permeation probability of H_2 is described as

$$\phi_P/\phi_{inc} = \frac{1}{2}\alpha_{H2}(C_O, T), \qquad (1)$$

where $\alpha_{H2}(C_O, T)$ is the sticking coefficient of H_2 molecules (the probability that a H_2 molecules is dissociatively absorbed by a tube at a single collision to the surface). According to the principle of detailed balance[6], $\alpha_{H2}(C_O, T) = k_r(C_O, T)\,K_S^2\sqrt{2\pi mkT}$ where K_S is the solubility of hydrogen in Ta, m is the mass of a H_2 molecule, k is the Boltzmann constant. The value of K_S reported in Ref. 9 was employed for the evaluation.

III. RESULTS AND DISCUSSION

III. A. Correlation between Bulk Oxygen Concentration and Permeation Probability

Under all conditions examined, the permeation rate (V) and desorption rate (Ta) were controlled by surface process and not by bulk diffusion process. Fig. 1 shows the correlation between bulk oxygen concentration and permeation probability at 700 ^{O}C as a typical example. The data for Nb obtained in a previous study[5] are also shown in this figure for comparison. The sharp reduction in the permeation probability with increasing bulk oxygen concentrations, C_O, was observed for all metals. Such

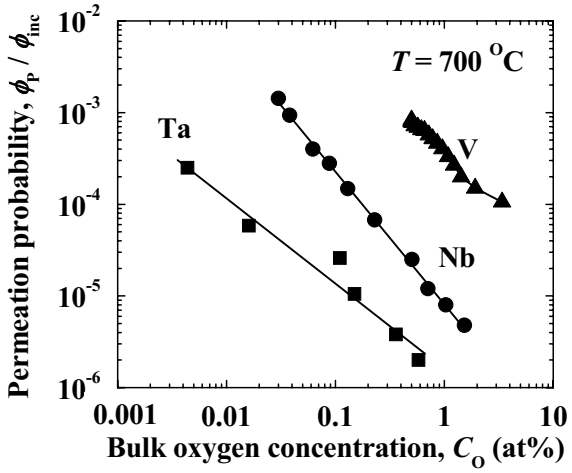


Fig. 1 Correlation between bulk oxygen concentration and permeation probability at 700 ^{o}C. Lines are guides to eyes.

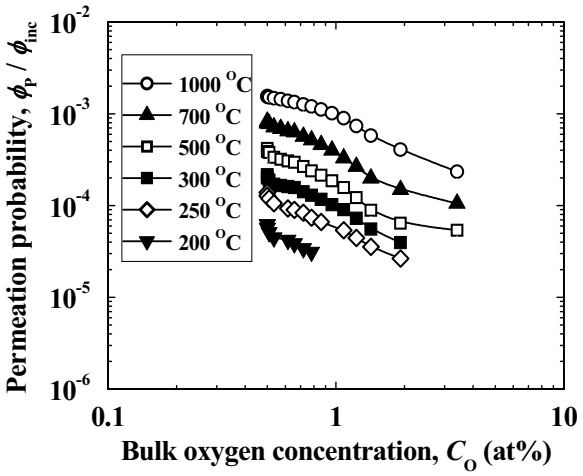


Fig. 2 Correlation between permeation probability and bulk oxygen concentration in V at various temperatures. Lines are guides to eyes.

reduction in permeation rate was ascribed to the increase in surface coverage of oxygen with increasing bulk oxygen concentration.[5,6,10] Significant difference, however, was observed in the sensitivity of permeation probability to oxygen. Interestingly, larger influence of oxygen on the permeation probability was observed for the metal with higher atomic number; the smallest values of permeation probability were obtained for Ta and the largest for V at given values of C_O.

Such a correlation between the influence of oxygen and atomic number held at all temperatures examined.

The permeation probability for V at various temperatures is plotted against C_O in Fig. 2. The reduction in ϕ_P/ϕ_{inc} with decreasing temperature is due to that in $\alpha_{H2}(C_O, T)$. Comparison with the similar plot for Nb shown in Fig. 5 of Ref. 6 clearly indicates the higher permeation probability for V than for Nb at all temperatures. Detailed surface analysis for V is under preparation to understand the mechanism underlying such weak influence of oxygen.

III. B. Permeation under Expected Blanket Conditions

In the vacuum permeator application, the upstream surfaces of permeation tubes face to Pb-17Li and tritium. The downstream surfaces are exposed to residual gas in vacuum system[1]; water vapor is one of the main components of residual gas in most vacuum systems. The temperature of Pb-17Li at the outlet of breeding zone is expected to be 700 ^{O}C in the dual coolant Pb–17Li breeder blanket concept.[1] The typical thickness of tubes is 0.5 mm.[1] The pressures of T_2 on both upstream and downstream sides of tubes could be changed with various parameters such as flow rate of Pb-17Li.

If vacuum condition on the downstream side is relatively poor and incident flux of water vapor and other oxygen-containing molecules is sufficiently large, so that the oxygen concentration in permeation tube material should be controlled by the partial pressures of the oxygen-containing gases. On the other hand, if the vacuum condition is sufficiently good, the oxygen concentration in tube material is controlled by the oxygen potential in Pb-17Li. According to Bhat and Borgstedt[11] who measured the oxygen potential in liquid Pb-17Li with various solid electrolyte oxygen meters, the solubility of oxygen in Pb-17Li is very small. This liquid alloy is saturated by oxygen in equilibrium with Li_2O in most cases. The values of oxygen potentials obtained with the various oxygen meters were close to the theoretical value evaluated by these authors from the standard Gibbs energy of formation of Li_2O and the activity of Li in Pb-17Li:

$$\overline{\Delta G_{O_2}} = -979785 + 278.79T, \qquad (2)$$

where $\overline{\Delta G_{O2}}$ is the oxygen potential.

The oxygen solubility in group 5 metals has been reported by Jehn and Fromm.[12] According to them, the entropy (ΔS_{sol}) and heat (ΔH_{sol}) of oxygen solution in group 5 metals are -101 J/Kmol O and -423 kJ/mol O for V, -96 J/Kmol O and -387 kJ/mol O for Nb, and -96 J/Kmol O and -383 kJ/mol O for Ta.

Fig. 3 shows the temperature dependence of oxygen partial pressure, P_{O2}, in Pb-17Li evaluated from $\overline{\Delta G_{O2}}$,

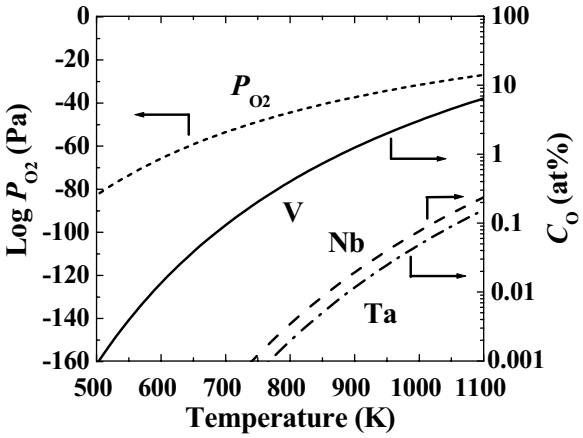


Fig. 3 Temperature dependence of oxygen partial pressure in Pb-17Li and corresponding bulk oxygen concentrations, C_O, in group 5 metals evaluated from data reported in Refs. 11 and 12.

and the bulk oxygen concentration, C_O, in group 5 metals at the corresponding temperature and oxygen partial pressure. Due to the more negative value of ΔH_{sol}, larger values of C_O were obtained for V than for Nb and Ta under the same condition. The oxygen concentration at 700 ^{O}C was calculated to be 2.4 at% for V, 0.054 at% for Nb and 0.034 at% for Ta. From the data shown in Fig. 1, the permeation probabilities at these oxygen concentrations were evaluated to be 1.3×10^{-4} for V, 5.5×10^{-4} for Nb, and 3.7×10^{-5} for Ta. Although the permeation probability for V showed the smallest sensitivity to oxygen, the value was significantly smaller than that for Nb due to the largest C_O in V. In other words, Nb should show the largest permeation rate under the conditions where oxygen concentrations in group 5 metals are controlled by the oxygen potential in Pb-17Li because of moderate oxygen solubility.

Vanadium may show the highest permeation rate if vacuum condition on the downstream side is relatively poor and consequently oxygen concentrations in the metals are much higher than the values in equilibrium with oxygen potential in Pb-17Li. The operation of a permeator under such conditions is, however, not preferable from the viewpoints of mechanical properties of group 5 metals and precipitation of Li_2O in Pb-17Li by oxygen permeation through group 5 metals. The present results were obtained after extensive sputter-cleaning as mentioned above; severe degradation in permeation rate by surface segregation of sulfur is possible for V if such treatment is not carried out.[13] By contrast, oxygen was observed on Nb surfaces as sole impurity in a wide temperature range without any treatment for impurity

removal.[14] These observations indicate that Nb is more suitable and controllable than V under the supposed blanket conditions.

Note that the results of present evaluation are sensitive to the values of ΔS_{sol} and ΔH_{sol}. Within the knowledge of the present authors, only Jehn and Fromm[12] have carried out these measurements. The accumulation of data on ΔS_{sol} and ΔH_{sol} is desirable to improve the reliability of this type of evaluation. Precise data on ΔS_{sol} and ΔH_{sol} are required also for the prediction of potential influence of oxygen on mechanical properties.

VI. CONCLUSIONS

The correlation between the surface reaction rates of hydrogen and bulk oxygen concentrations were examined for V and Ta and compared with that for Nb (Ref. 5) to understand the potential influence of oxygen on the tritium recovery from Pb-17Li by vacuum permeation. The surface reaction rates of hydrogen on these metals, and consequently permeation rates, decreased with increasing bulk oxygen concentrations. The weakest influence of oxygen on the hydrogen permeation rate was observed for V. The expected permeation rate through V under typical blanket conditions, however, was not necessarily high due to high oxygen solubility in V. The evaluation indicated that the highest permeation rate should be obtained with Nb under typical blanket conditions.

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